

SVM:

Support vector machine

* Support vector machine :

SVM is a supervised machine learning algorithm we give it labeled data of two classes, and it learns the best boundary that separates them. This boundary is called the hyperplane.

SVM is important because

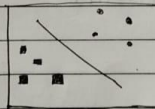
- It does not pick just any boundary. It picks the safest one, which stays far from both classes.
- It works very well when data has many features.
- It needs a few special points to make its decision, so the model is small and fast.

* famous Real ex :

Email spam detection decides if a mail is spam or not.

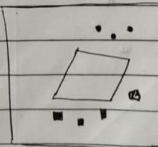
- Hyperplane : The hyperplane is the boundary that separates the classes. Its shape depends on how many features the data has.

(i) 2D data (2 features) :



hyperplane = a line

(ii) 3D data (3 features)

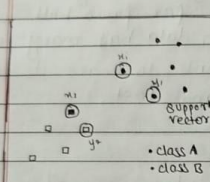


hyperplane = a plane

(iii) 4D data and above : we cannot draw it, but the same idea continues that is why it is called a hyperplane

Algorithm

- Step 1: Find the support vectors
- Take two classes, class A and class B
 - From each class, find the points that are closest to the other class
 - These border points are called support vectors
 - They are the most difficult points, and they alone decide the boundary



Q. how to find them?

→ Measure the distance betn point of opposite classes.

The pairs with the smallest distance gives the support

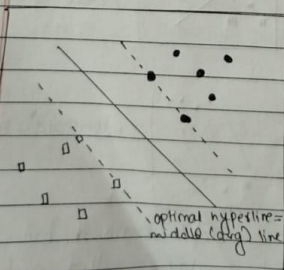
vector: $d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

Step 2: Draw a boundary line through the support vector of each class

Step 2: Take the Average Line

→ Draw the line exactly in the middle of the two boundary lines. This is average line

→ The average line is our final answer, the optimal hyperplane



Q. what does the avg line tell us?

→ It is equally far from both classes, so it gives the biggest possible margin. Any new points is classified by checking which side of this line it falls on.

→ A kernel is a

A kernel is a trick used in SVM to handle non-linear data.

If a line cannot separate the data, a kernel lifts the data to a higher dimension, like $f(x) = x^2$ and then the same steps work.

Hinge loss :-

Hinge loss is a loss function used in SVM to measure classification error.

Calculate hinge loss for all data points and add them.

$$\text{Total loss} = L(P_1) + L(P_2) + \dots + L(P_n)$$

Minimum Hinge loss :-

- line moves toward class A $\rightarrow$ A's losses increase
- line moves toward class B $\rightarrow$ B's losses increase
- line in the middle $\rightarrow$ total loss is minimum

Minimum hinge loss = Max Margin line = the optimal hyperplane.

KNN:

* K Nearest Neighbours *

KNN is a

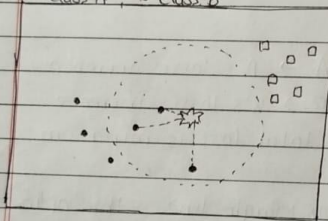
KNN is a supervised algorithm that classifies a new points by looking at its k closest existing points and taking a majority vote.

→ KNN is called lazy learner: it does no training at all. It just store the data and work only when a new points arrives.

→ It can do classification (maj vote) and regression (avg. of neighbours)

Q: How KNN works:

• class A, • class B



→ choose a value of k (3)

→ Measure the distance from a new point to every training point.

→ Pick the k -closest points

→ classification: take maj vote of their classes

→ Regression: take a avg.

$k=3$ nearest neighbours

votes: 3 × class A, 0 × class B

At where is KNN used?

→ Recommendation systems find user or items most similar to you.

→ Medical diagnosis compare a patient to similar past patients

→ Search and matching find the most similar documents products.

one sentence to remember:-

Any problem of the type "find the most similar things" is a nearest neighbour problem.

→ KNN is ~~used~~ preferred when data is small, pattern are not linear, and predictions can be made based on similarity betⁿ data points.

- classification (Predicting class)

KNN

Example Predicting Movie genre

IMDb Rating	Duration	Genre
8.0 (Mission impos)	160	Action
6.2 (Gadar 2)	170	Action
7.2 (Rocky & Rani)	168	comedy
8.2 (OMG 2)	155	comedy

Now predict the genre of Bashir with IMDb rating 7.4 and duration 114 minutes

$$x_1, y_1 = (7.4, 114)$$

Calculate Distance by Euclidean Distance

$$\begin{aligned}\text{Distance to } (8.0, 160) &= \sqrt{(7.4 - 8.0)^2 + (114 - 160)^2} = 46.60 \\ (6.2, 170) &= \sqrt{(7.4 - 6.2)^2 + (114 - 170)^2} = 56.01 \\ (7.2, 168) &= \sqrt{(7.4 - 7.2)^2 + (114 - 168)^2} = 54.0 \\ (8.2, 155) &= \sqrt{(7.4 - 8.2)^2 + (114 - 155)^2} = 41.0\end{aligned}$$

let $k=3$

Select k nearest neighbour

Action, comedy, comedy

By classification voting we conclude it's a comedy movie